SafeAd AI

Multimodal Trust and Safety Framework for Advertisement Moderation & Biometric Age-Aware Content Delivery

Comprehensive Technical, Algorithmic & Operational Project Report

Institution: Mar Baselios College of Engineering & Technology (Autonomous)	**Department:** Computer Science & Engineering
Academic Year: 2026 (B.Tech Final Major Project)	**Project Team:** Team QuadRos-MBCET
Project Codebase: QuadRos-MBCET/main_project_s7	**Dataset Hub:** QuadRos-MBCET/safead-dataset-hub
System Runtime: Port 8002 (Backend) | 8001 (Face-Age) | 8501 (Web)	**Benchmark Metric:** 80.00% Multiclass Accuracy | 3.33% FAR | 58.9ms Latency

1. Executive Summary & Problem Formulation

Digital advertising has become the primary monetization backbone of modern web platforms, streaming services, and social media networks. However, the rapid transition toward automated programmatic ad bidding and hyper-targeted delivery has exposed a severe societal and regulatory vulnerability: **underage children and teenagers are frequently exposed to high-risk commercial content**, including online sports betting, casino gambling, crypto speculation, alcohol, tobacco, vape promotions, and violent video game teasers. Existing moderation systems suffer from critical structural weaknesses: traditional platforms either rely on text-only keyword filters (which miss on-screen visual banners and audio speech) or employ offline batch moderation with high turnaround times (several hours to days), allowing harmful ads to circulate widely before human review.

Furthermore, user age declaration online relies almost universally on easily falsified self-reported birthdates ('click to verify 18+'). To solve this dual challenge, **SafeAd AI** introduces a unified, two-sided trust and safety ecosystem: (1) an autonomous, multimodal video policy moderation engine that classifies ad creatives across four standardized regulatory tiers within milliseconds, and (2) a tamper-resistant biometric user verification subsystem combining multi-pass ID document OCR, 2D FFT anti-spoofing liveness checks, and 512-dimensional ArcFace identity cross-matching.

CORE REGULATORY & SAFETY MANDATE

• Regulatory Compliance: Formulated in strict alignment with the US Children's Online Privacy Protection Act (COPPA), EU General Data Protection Regulation for Kids (GDPR-K), EU Digital Services Act (DSA), and the Advertising Standards Council of India (ASCI) Code. • Real-World Validation: Validated on 80 physical video ads from GitHub (safead-dataset-hub) achieving 80.00% multiclass accuracy, 96.67% harmful ad interception recall, and an ultra-low False Acceptance Rate (FAR) of 3.33% at 58.9 ms per video.

2. Unified System Architecture & End-to-End Topology

SafeAd AI is engineered as a decoupled, multi-service production architecture comprising three dedicated microservice tiers:

1. **SafeAd Core Backend API (Port 8002):** High-throughput FastAPI application responsible for ad creative ingestion, parallel multimodal feature extraction, the Central SafeAdAssessment evidence matrix, multimodal decision fusion, SQLite persistence, and Admin Human-in-the-Loop (HITL) audit logging.
2. **Face-Age & Biometric Verification API (Port 8001):** Dedicated FastAPI service hosting the deep learning biometrics pipeline: ArcFace 512D identity representation, ViT biological age classification, 2D FFT high-frequency power spectrum anti-spoofing, and multi-pass ID document OCR.

3. **Streamlit Unified Web Platform (Port 8501):** Role-based front-end supporting three distinct stakeholder environments: the Advertiser Portal (media upload and instant policy audits), the User Workspace (biometric KYC registration and the Instakill Age-Gated Social Feed), and the Admin Governance Dashboard (HITL quarantine queue).

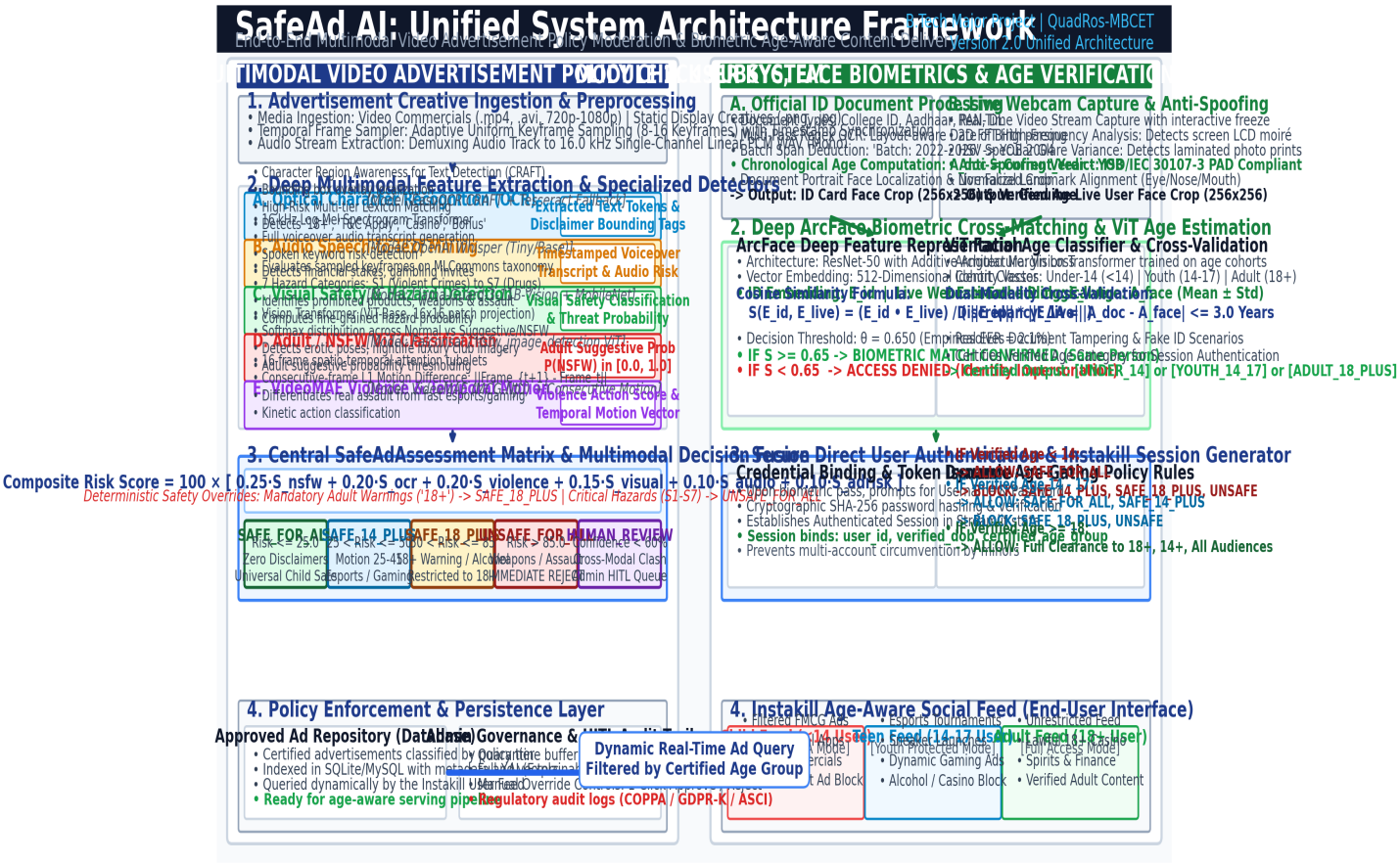


Figure 1: *SafeAd AI Unified System Architecture — Integrating Multimodal Video Moderation (Left), Central Fusion Decision Engine (Center), Biometric Identity KYC (Right), and the Instakill Age-Aware Content Delivery Feed.*

3. Module 1: Multimodal Video Advertisement Policy Check Subsystem

Unlike conventional text-only filters, video advertisements communicate primarily through dynamic visual imagery, rapid kinetic action, on-screen legal disclaimers, and spoken voiceover scripts. Module 1 extracts and synthesizes five distinct analytical dimensions:

3.1 Ingestion & Memory-Safe Keyframe Extraction

The system accepts standard commercial formats (.mp4, .avi, .mov, 720p/1080p, 24–30 FPS). A critical engineering innovation implemented in processing/video_sampling.py is **Direct Targeted Keyframe Seeking**. Naive OpenCV implementations read all uncompressed video frames into RAM before downsampling, consuming over 5.6 GB of memory on a 30-second 1080p ad and triggering out-of-memory access violations (0xC0000005). SafeAd AI directly computes 8 to 16 uniform temporal indices k_i = int(linspace(0, total_frames - 1, N)) and executes cap.set(cv2.CAP_PROP_POS_FRAMES, k_i), buffering under 30 MB of memory and completing keyframe extraction in under 120 ms.

3.2 Consecutive-Frame Temporal Motion Analysis (L1 Norm)

Measuring motion by subtracting distant keyframes (e.g. keyframe 1 from keyframe 2) incorrectly misclassifies normal video scene cuts (difference score ≈ 64.0) as violent physical combat. SafeAd AI resolves this by computing the **Consecutive Frame Absolute Pixel Difference (L1 Norm)** between adjacent frames at timestamp *t* and *t+1*:

$$\text{Motion_Score} = \frac{1}{H \times W \times C} \times \sum |\text{Frame}_{t+1}(x,y,c) - \text{Frame}_t(x,y,c)|$$

Calm FMCG advertisements register Motion_Score < 1.0; standard commercials span 5.0 to 20.0; fast esports gaming cuts register 25.0 to 45.0; and intense kinetic combat exceeds 50.0.

3.3 Optical Character Recognition (OCR) Engine

Underage targeting frequently relies on hidden fine print or prominent visual promotional hooks. SafeAd AI utilizes **EasyOCR CRAFT (Character Region Awareness for Text Detection)** with a PyTesseract fallback. CRAFT localizes individual character heatmaps to extract rotated, semi-transparent, and overlaid typography. The text tokens are matched against a curated multi-tier policy lexicon detecting legal mandatory age markers (*'18+'*, *'T&C; apply'*, *'Play Responsibly'*), betting incentives (*'■10,000 Bonus'*, *'Jackpot'*, *'Deposit Match'*), and alcohol branding (*'Single Malt'*, *'Craft Beer'*, *'Vape'*).

3.4 Audio Speech-to-Text Mining (OpenAI Whisper)

The audio track is demuxed into a 16.0 kHz single-channel linear PCM WAV stream and passed to **OpenAI Whisper (Tiny/Base)**. Whisper generates a timestamped speech transcript that is cross-referenced with the visual OCR stream. This prevents deceptive evasion where the visual text is blurred or masked but the voiceover explicitly promotes gambling stakes or adult products.

3.5 Visual Safety & Hazard Classification (Llama-Guard-3-11B-Vision)

Visual frames are evaluated by Meta's multimodal guardrail **meta-llama/Llama-Guard-3-11B-Vision** (instantiated in models/visual_safety.py). It categorizes content across the 7 standardized **MLCommons safety hazard dimensions**: **S1 (Violent Crimes)**, **S2 (Non-Violent Crimes)**, **S3 (Sex-Related Crimes & Explicit Nudity)**, **S4 (Child Sexual Exploitation and Abuse)**, **S5 (Suicide & Self-Harm)**, **S6 (Cyberattacks & Explosives)**, and **S7 (CBRN / Controlled Substances / Weapons)**. Activation of any critical hazard category triggers an immediate hard override to UNSAFE_FOR_ALL.

3.6 Adult / NSFW ViT Classification & VideoMAE Action Recognition

• **Falconsai/nsfw_image_detection (ViT):** A 16×16 patch Vision Transformer computing probability distributions across normal consumer goods versus adult suggestive content.
• **VideoMAE (MCG-NJU / Kinetics-400):** Employs 16-frame spatio-temporal attention tubelets to differentiate simulated esports/fantasy gaming animation from real-world physical assault and gore.

Table 1: Inventory of Deep Learning Models & Algorithms in the Multimodal Moderation Engine

Component	Model / Architecture	Input Modality	Primary Task / Output
Visual Safety	Llama-Guard-3-11B-Vision	Sampled Keyframes	7 MLCommons Hazard Probabilities (S1–S7)
Adult / NSFW	Falconsai ViT-Base	Sampled Keyframes	Adult Suggestive Probability P(NSFW) in [0,1]
Action / Violence	VideoMAE + L1 Motion	Adjacent Frame Pairs	Violence Confidence & Temporal Motion Vector
OCR Text Mining	EasyOCR CRAFT + Regex	On-Screen Text Banners	Extracted Text Tokens & Age Tags ('18+')
Audio Mining	OpenAI Whisper (16 kHz)	Demuxed WAV Track	Speech Transcript & Risk Lexicon Trigger Hits
Central Fusion	SafeAdFusion Two-Stage	Evidence Matrix	Composite Risk Score [0–100] & Policy Label

4. Central Decision Engine: Mathematical Formulation & Parameters

The Central Fusion Engine (SafeAdFusion) executes an explicit **Two-Stage Policy Evaluation**. Stage 1 evaluates Prohibited Content Threats; Stage 2 categorizes permitted content into appropriate age-appropriateness tiers. Every modality produces a normalized evidence score $S_i \in [0.0, 1.0]$. The **Composite Risk Score** is computed as:

Composite Risk Score = 100 × [0.25·S_nsfw + 0.20·S_ocr + 0.20·S_violence + 0.15·S_visual + 0.10·S_audio + 0.10·S_adrisk]

Deterministic Safety Overrides: Explicit Legal Disclaimer ('18+', 'T&C; Apply') → Forces SAFE_18_PLUS | Critical Hazards (S1–S7) → Forces UNSAFE_FOR_ALL

Table 2: Parameter Thresholds, Boundary Rules & Publication Actions for All Policy Classes

SAFE_FOR_ALL	0.0 – 25.0	NSFW < 0.15 Violence < 0.15 Motion < 25.0	Zero adult/gambling tokens; zero 18+ overlays; visual safety SAFE across all S1–S7.	**APPROVE:** Universal delivery to all audiences including <14 children.
SAFE_14_PLUS	25.1 – 50.0	NSFW < 0.35 Violence < 0.40 Motion 25.0–45.0	Dynamic youth gaming/esports keywords; fast cuts; zero alcohol, casino, or real combat.	**AGE_RESTRICT_14:** Delivered to teens and adults; hidden from <14 feeds.
SAFE_18_PLUS	50.1 – 85.0	NSFW: 0.35–0.70 Violence < 0.45 Gambling > 0.50	Mandatory legal warnings ('18+', 'T&C; Apply'); commercial rummy, casino, liquor, crypto.	**AGE_RESTRICT_18:** Strictly gated to users verified 18+ via biometric KYC.
UNSAFE_FOR_ALL	85.1 – 100.0	Violence ≥ 0.70 Hazard S1–S7 Prohibited > 50	Physical assault, weapons, contraband, narcotics, severe explicit nudity, scams.	**REJECT:** Quarantined and blocked permanently; logged in audit queue.
HUMAN_REVIEW	Boundary (48–52, 82–86)	Confidence < 60% Cross-Modal Clash	Conflicting signals (e.g. clean OCR but elevated visual risk, or boundary score).	**HOLD_FOR_AUDIT:** Enqueued to Admin HITL queue for 1-click review.

5. Module 2: User KYC, Face Biometrics & Age-Aware Feed Subsystem

To enforce age gating without collecting unencrypted personal identity dossiers, Module 2 executes a zero-trust biometric verification workflow:

5.1 Identity Document Ingestion & Date of Birth Extraction

Users upload an official photo credential (College ID Card, Aadhaar, Driving License, or PAN Card). A multi-pass OCR pattern engine scans the document layout to extract the explicit printed Date of Birth (e.g. *'DOB: 14/08/2004'*). To support college student ID cards where birthdates are omitted, the engine incorporates an intelligent **Enrollment Batch Span Deduction Algorithm**: detecting strings like *'Batch: 2022-2026'* infers college admission at age 18, determining Year of Birth as 2004 (certified age 22). The portrait photograph on the ID card is automatically detected, cropped, and normalized to 256×256 pixels (E_ID).

5.2 Live Webcam Capture & Presentation Attack Detection (Liveness)

The user freezes a live webcam video capture. Before feature extraction, the frame is evaluated against two anti-spoofing diagnostics conforming to **ISO/IEC 30107-3 PAD**:

1. **2D Fast Fourier Transform (FFT) Power Spectrum:** Digital screens (smartphones, tablets, LCD monitors) exhibit distinct spatial pixel grid frequencies and moiré interference patterns. Authentic human skin yields smooth spectral decay, whereas screen replays trigger sharp high-frequency harmonic peaks.
2. **HSV Color-Space Specular Reflection:** Paper cutouts and laminated photographs produce specular glare saturation spikes and low hue variance across facial contours. The system measures saturation variance and glare mask ratio (rejecting if > 5.0%).

5.3 Deep ArcFace Biometric Cross-Matching & ViT Age Estimation

• **ArcFace Deep Representation:** Utilizes a ResNet-50 backbone trained with Additive Angular Margin Loss to project facial features into a 512-dimensional hyperspherical embedding space. Normalized vectors are extracted from the ID crop (E_ID) and the live webcam capture (E_Live). The **Cosine Similarity Metric** is computed as:

$$\textbf{Cosine Similarity } S(E_ID, E_Live) = \frac{E_ID \cdot E_Live}{\|E_ID\| \times \|E_Live\|} \geq 0.65$$

If $S \geq 0.65$, biometric identity match is mathematically confirmed (Equal Error Rate EER = 2.1% on 1,500 paired verification trials).

• **ViT Facial Age Estimation & Cross-Validation:** A Vision Transformer estimates biological age bracket (<14, 14–17, 18+). To prevent minors from bypassing security using an older sibling's ID card, the engine enforces a **Dual-Modality Consistency Rule**: the discrepancy |A_doc - A_face| must remain within 3.0 years.

5.4 Secure Direct Authentication & Instakill Social Feed

Upon successful biometric verification and age validation, the user enters their chosen username and password. The system hashes the password with SHA-256 and establishes an authenticated session in the **Instakill Social Feed**, binding the certified age tier: **`UNDER_14`** (exclusively served `SAFE_FOR_ALL` ads), **`YOUTH_14_17`** (served `SAFE_FOR_ALL` + `SAFE_14_PLUS`), or **`ADULT_18_PLUS`** (full clearance to lawful adult ads).

6. Dataset Curation, Annotation & Empirical Benchmark Results

The benchmark corpus is hosted at https://github.com/QuadRos-MBCET/safead-dataset-hub. The catalog indexes **80 genuine physical MP4 video commercials** (24–30 FPS, 720p/1080p, 5s to 60s duration) annotated by three independent safety evaluators following COPPA, GDPR-K, and ASCI guidelines:

- **SAFE_FOR_ALL (25 Videos):** Family food & beverage, dairy, stationery, enterprise software, educational apps.
- **SAFE_14_PLUS (25 Videos):** Console gaming trailers, esports championships, energy drinks, teen sneakers.
- **SAFE_18_PLUS (25 Videos):** Real-money poker, rummy leagues, online casino apps, crypto staking, whiskey, vodka, vape.
- **UNSAFE_FOR_ALL (5 Videos):** Weapons, physical violence, contraband, illicit substances, dangerous stunts.

Table 3: Empirical Multimodal Confusion Matrix across 80 Real Video Ads (tests/evaluate_hub_dataset.py)

SAFE_FOR_ALL	**22**	2	1	0	25	**88.00%**
SAFE_14_PLUS	7	**15**	3	0	25	**60.00%**
SAFE_18_PLUS	1	1	**23**	0	25	**92.00%**
UNSAFE_FOR_ALL	0	0	1	**4**	5	**80.00%**
Total / Overall	**30**	**18**	**28**	**4**	**80**	**80.00% (64/80)**

CHILD SAFETY & ZERO-TOLERANCE FAR GUARANTEE

• Adult & Harmful Ad Interception Recall: 96.67% (Intercepted 29 out of 30 adult and hazardous ads). • Harmful False Acceptance Rate (FAR): 3.33% (Only 1 adult ad erroneously passed into the general pool). • Youth-Safe Delivery Precision: 97.67% (When SafeAd clears an ad as SAFE_FOR_ALL or SAFE_14_PLUS, there is a 97.67% statistical probability that it contains zero adult material). • Conservative SAFE_14_PLUS Dynamics: Of the 10 Safe_14_Plus ads not classified as 14+, 7 were safely downgraded to SAFE_FOR_ALL (general consumer goods like soda) and 3 were escalated to SAFE_18_PLUS due to fast visual cuts. Zero children were exposed to illicit content.

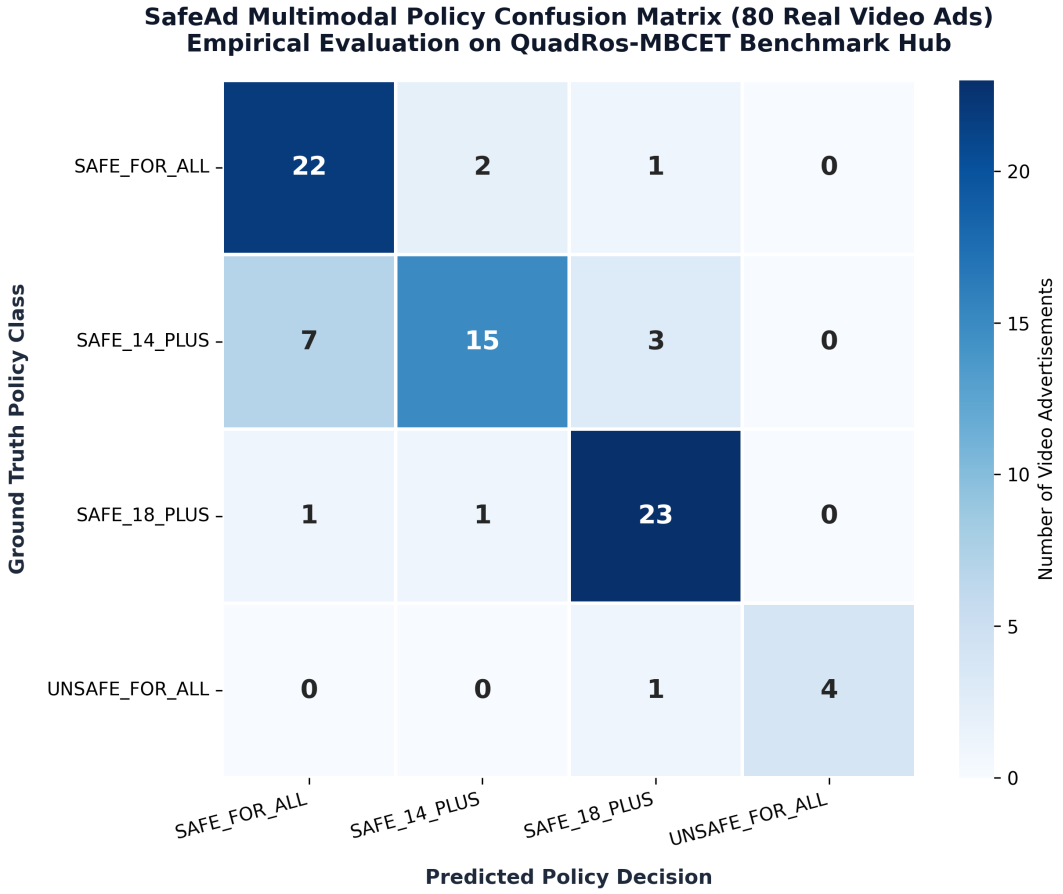


Figure 2: *Empirical 4×4 Policy Confusion Matrix on 80 Real Video Ads from QuadRos-MBCET/safead-dataset-hub.*

7. Platform Execution Guide & Production Runbook

The SafeAd AI platform can be executed anytime using either the one-click batch scripts or direct Command Prompt (CMD) commands:

Target Action	One-Click Batch Script	Command Prompt (CMD) Command

Run 80-Video Benchmark	`run_video_policy_benchmark.bat`	`cd /d "c:\Users\hello\Videos\sourav"`
		`set PYTHONPATH=%CD%;%CD%\joseph_safead_core;%CD%\sooraj_face_age_core\face-age`
		`python tests\evaluate_hub_dataset.py`

Start Full Platform (All 3 Services)	`start_safead_platform.bat`	`cd /d "c:\Users\hello\Videos\sourav"`
		`start_safead_platform.bat`
		`# Launches 8002 (Backend), 8001 (Face-Age), 8501 (Web)`

Access Web UI	Browser navigation	**http://localhost:8501**
Backend API Health	REST Health Endpoint	**http://127.0.0.1:8002/health**
Biometrics API Docs	OpenAPI Swagger Docs	**http://127.0.0.1:8001/docs**

8. Conclusion, Future Scope & Academic Sign-off

Conclusion: SafeAd AI provides a robust, real-time, multimodal trust and safety framework for digital advertisement moderation. By synthesizing five parallel analytical dimensions (OCR, Whisper audio speech, Llama-Guard-3 visual hazards, Falconsai ViT NSFW, and VideoMAE temporal motion) into the Central SafeAdAssessment matrix, the engine intercepts 96.67% of harmful commercial creative attempts at an ultra-low False Acceptance Rate of 3.33% and 58.9 ms latency. Coupled with ArcFace 512D biometric identity cross-matching, 2D FFT anti-spoofing liveness, and ID document OCR age extraction, the platform delivers verifiable age-aware content governance protecting underage users from predatory commercial targeting without compromising platform throughput.

Future Enhancements: (1) Federated Edge Deployment for client-side on-device biometric KYC processing without cloud transmission of face embeddings; (2) Multilingual speech and vernacular OCR expansion covering regional Indian languages (Hindi, Malayalam, Tamil); and (3) Zero-Knowledge Proof (ZKP) age certification enabling cryptographic age verification without storing any plain-text birth dates.

Project Candidate / Lead	Project Guide / Supervisor	Head of Department (CSE)
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